

TASK 1 : Install Docker in ubuntu VM ,pull docker image nginx and run container from the image

Vm created

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Create a virtual machine

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Instance details

Virtual machine name *

Region *

Availability options

Zone options ☒ Self-selected zone
Choose up to 3 availability zones, one VM per zone

☐ Azure-selected zone (Preview)
Let Azure assign the best zone for your needs

Availability zone *

☒ You can now select multiple zones. Selecting multiple zones will create one VM per zone. [Learn more](#)

Security type

Image *

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CreateVm-canonical.ubuntu-24_04-lts-server-20260210131233 | Overview

Deployment

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Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name: CreateVm-canonical.ubuntu-24_04-lts-server-2... Start time: 2/10/2026, 1:14:49 PM

Subscription: Vanguard Class Training Correlation ID: 6d573363-e429-4e6b-9d21-54cccd68c120

Resource group: FardeenAttar-rg

Deployment details

Next steps

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- Installed Docker on ubuntu VM
- Pulled nginx

```
demo@demovm-docker:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
nginx:latest	341bf0f3ce6c	240MB	65.8MB	

```
demo@demovm-docker:~$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
bae5a1799a80: Pull complete
4f4efe02d542: Pull complete
0c8d55a45c0d: Pull complete
46bf3a120c8e: Pull complete
f73400a233fd: Pull complete
7b6cb8ccac7b: Pull complete
47cd406a84ef: Pull complete
2e02dba24409: Download complete
a5d78d617315: Download complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
demo@demovm-docker:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
nginx:latest	341bf0f3ce6c	240MB	65.8MB	

- Ran the container by `docker run -dt -p 8080:80 nginx`

```
demo@demovm-docker:~$ docker run -dt -p 8080:80 nginx
199b4eb3883bc53f7c42e57ee11f874ed558a25e3a95ab16d0b8b7fd42e89c4a
demo@demovm-docker:~$ cd /usr/share/nginx/html/
```

- Logged in the container, Installed `vim` editor in container (By default not installed)

```
demo@demovm-docker:~$ docker exec -it 199b4eb3883bc /bin/bash
root@199b4eb3883b:/# ls
```

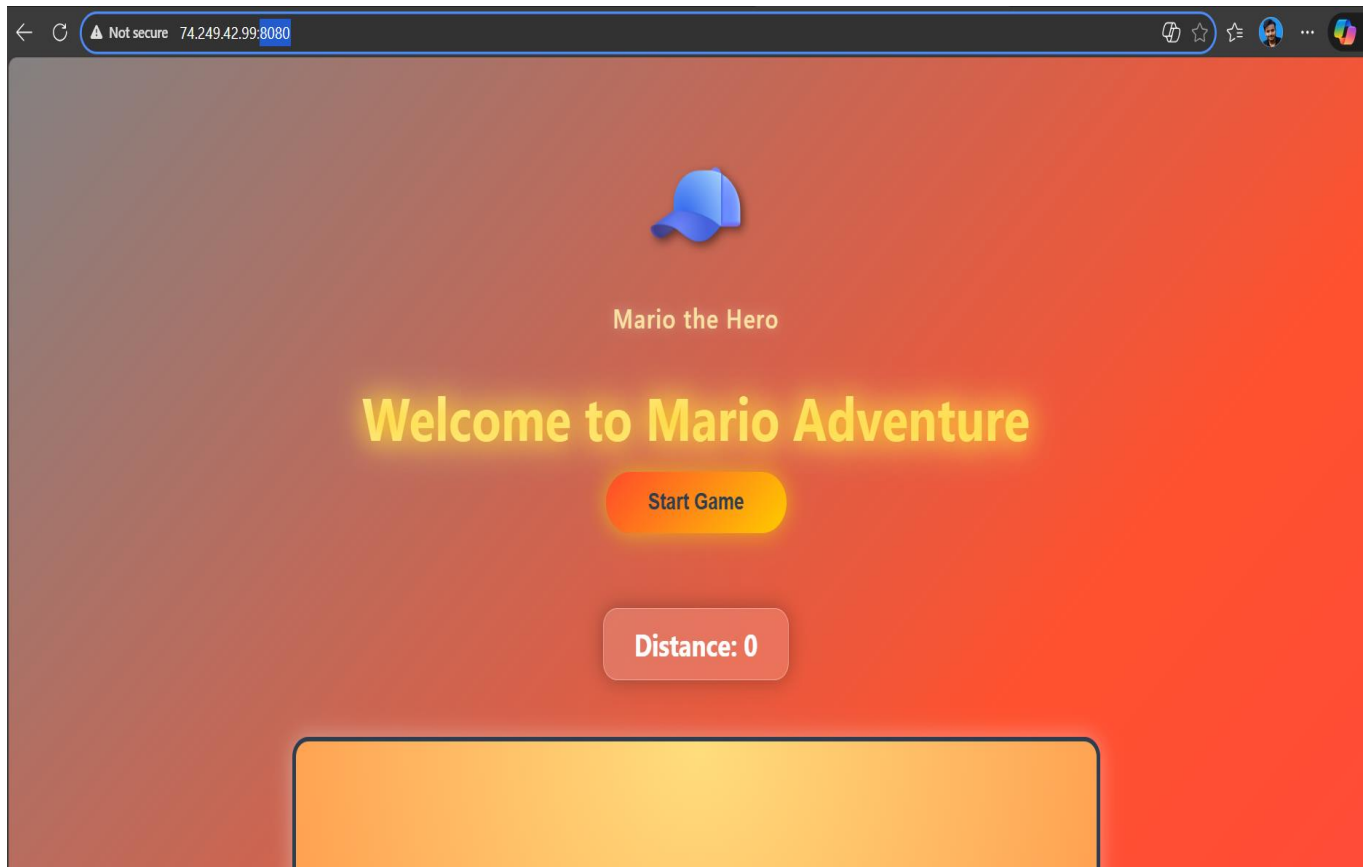
```
root@199b4eb3883b:/usr/share/nginx/html# apt install -y vim
Installing:
vim

Installing dependencies:
libgpm2 libsodium23 vim-common vim-runtime xxd
```

- Added custom `index.html` and `main.js` at `/usr/share/nginx/html/index.html` `main.js` Which is sample mario game
- After edit exited from container `ctrl + aq`

```
root@199b4eb3883b:/usr/share/nginx/html# vi index.html
root@199b4eb3883b:/usr/share/nginx/html# vi main.js
```

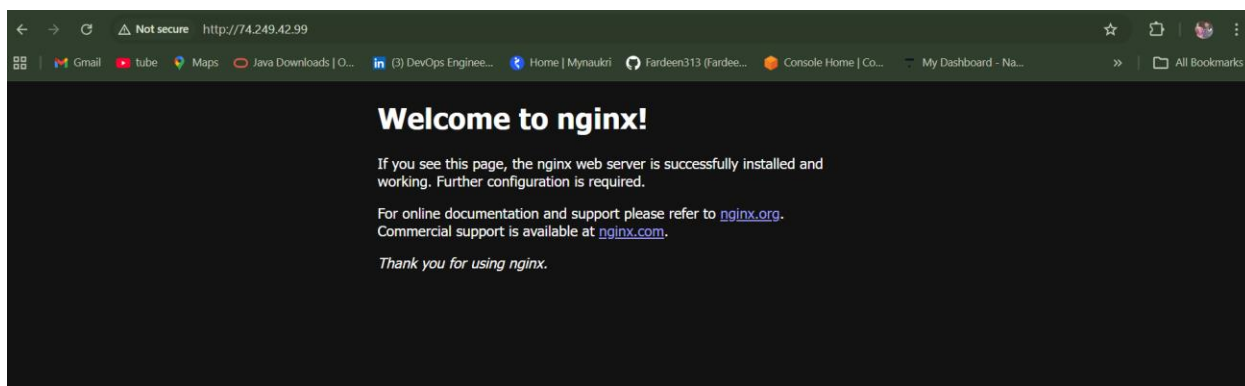
- Then accessed it at port `<vm-ip>:8080`



TASK 2 : Create A Docker volume and create a container from that volume

- Created Docker Volume
- Ran the nginx container from the volume so even if container is deleted we have container files and folders inside volume

```
demo@demovm-docker:~$ docker volume create nginx_data
nginx_data
demo@demovm-docker:~$ ls
nginx-site
demo@demovm-docker:~$ docker volume ls
DRIVER      VOLUME NAME
local       nginx_data
demo@demovm-docker:~$ docker run -d \
-p 80:80 \
-v nginx_data:/usr/share/nginx/html \
nginx
1dae9e55f4ea8d180d11916b40cfb5929249f06c5dbe98482dfb2710934aa404
demo@demovm-docker:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                                                                 NAMES
1dae9e55f4ea   nginx    "/docker-entrypoint.    5 seconds ago Up 4 seconds   0.0.0.0:80->80/tcp, [::]:80->80/tcp   practical_payne
651b3950099f   nginx    "/docker-entrypoint.    12 minutes ago Up 12 minutes   0.0.0.0:3000->80/tcp, [::]:3000->80/tcp   stupefied_villani
74d651a00ed2   nginx    "/docker-entrypoint.    22 minutes ago Up 22 minutes   0.0.0.0:8090->80/tcp, [::]:8090->80/tcp   relaxed_lovelace
199b4eb3883b   nginx    "/docker-entrypoint.    36 minutes ago Up 36 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   naughty_shamir
demo@demovm-docker:~$ docker volume inspect nginx_data
[
  {
    "CreatedAt": "2026-02-10T09:20:31Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/nginx_data/_data",
    "Name": "nginx_data",
    "Options": null,
    "Scope": "local"
  }
]
```



TASK 3 : Create Custom webserver based **Dockerfile** ,build an image , run the container and push it into Dockerhub repository and Azure container registry

- Created Dockerfile
- Built image webapp

```
demo@demovm-docker:~$ vi Dockerfile
demo@demovm-docker:~$ vi index.html
demo@demovm-docker:~$ docker build -t .
ERROR: docker: 'docker buildx build' requires 1 argument

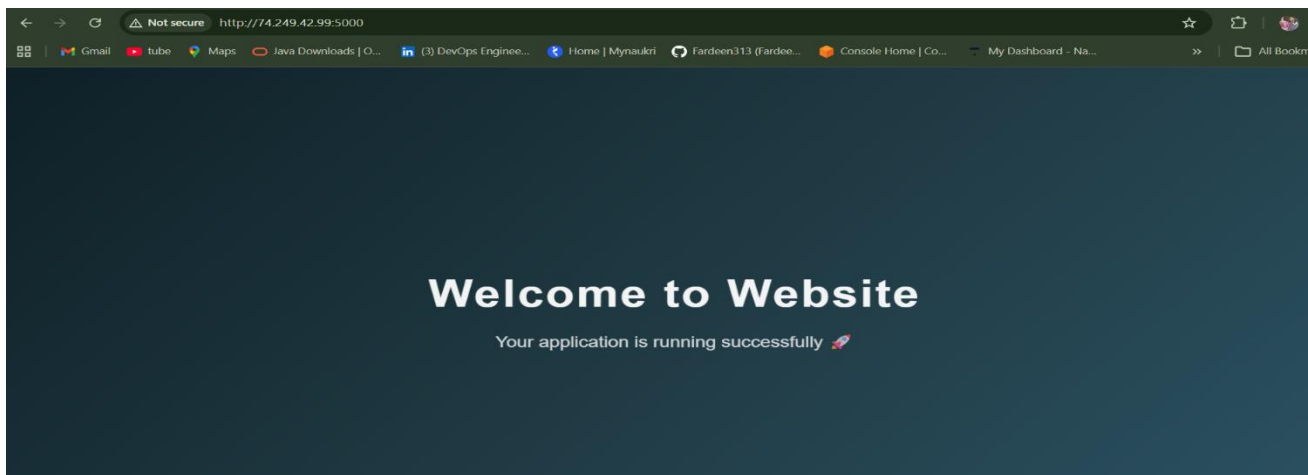
Usage:  docker buildx build [OPTIONS] PATH | URL | -

Run 'docker buildx build --help' for more information
demo@demovm-docker:~$ docker build -t webapp .
[+] Building 1.9s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 142B
=> [internal] load metadata for docker.io/library/nginx:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 1.18kB
=> [1/2] FROM docker.io/library/nginx:latest@sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
=> => resolve docker.io/library/nginx:latest@sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
=> [2/2] COPY index.html /usr/share/nginx/html/index.html
=> exporting to image
=> => exporting layers
```

- Ran the container at 5000 port

```
demo@demovm-docker:~$ docker run -dt -p 5000:80 fardeenattar/webapp:latest
6b14681346dc1e9a78b362a9d3f2620f36c162e758f74c66fffb520ab0469b
```

- Accessed it via browser



- Pushed Into **Dockerhub** And **Azure Container registry (ACR)**

```
demo@demovm-docker:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
nginx:latest	341bf0f3ce6c	240MB	65.8MB	U
webapp:latest	bcfbac3231ca	237MB	62.9MB	

```
demo@demovm-docker:~$ docker tag webapp:latest fardeenattar/webapp:latest
demo@demovm-docker:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
fardeenattar/webapp:latest	bcfbac3231ca	237MB	62.9MB	
nginx:latest	341bf0f3ce6c	240MB	65.8MB	U
webapp:latest	bcfbac3231ca	237MB	62.9MB	

```
demo@demovm-docker:~$ docker push fardeenattar/webapp:latest
The push refers to repository [docker.io/fardeenattar/webapp]
0c8d55a45c0d: Mounted from library/nginx
7b6cb8ccac7b: Mounted from library/nginx
f73400a233fd: Mounted from library/nginx
bae5a1799a80: Mounted from library/nginx
33be21097423: Pushed
a96de523845a: Pushed
46bf3a120c8e: Mounted from library/nginx
4f4efe02d542: Mounted from library/nginx
47cd406a84ef: Mounted from library/nginx
latest: digest: sha256:bcfbac3231ca95a3c55c10e35dfecd197f31860a239593c3357c2ae14e1f0351 size: 856
```

hub Explore My Hub Search Docker Hub CtrlK

Explore / fardeenattar / webapp



fardeenattar/webapp

By fardeenattar · Created less than a minute ago

☆0 ↓0

Manage Repository

Overview Tags

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- Pushed Into **Azure Container registry (ACR)**
- To push first need to install Azure CLI inside VM
- Then Az login – tenant <id>
- Create Container registry

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Home > Container registries

Create container registry

...

Basics

Networking

Encryption

Tags

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Azure Container Registry allows you to build, store, and manage container images and artifacts in a private registry for all types of container deployments. Use Azure container registries with your existing container development and deployment pipelines. Use Azure Container Registry Tasks to build container images in Azure on-demand, or automate builds triggered by source code updates, updates to a container's base image, or timers. [Learn more](#)

Project details

Subscription *

Vanguard Class Training

Resource group *

FardeenAttar-rg

Create new

Instance details

Registry name *

webserver

✓

.azurecr.io

Location *

Austria East

Domain name label scope * ⓘ

Unsecure

Registry domain name

webserver.azurecr.io

Use availability zones ⓘ

☐

Availability zones are activated on premium registries and in regions that

Review + create

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Next: Networking >

- Then az acr login
- Then Tag and push image

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
demo@demovm-docker:~$ az login --tenant 6d7e0652-b03d-4ed2-bf86-f1999cecd17
To sign in, use a web browser to open the page https://microsoft.com/devicelogin and enter the code AZNVPT6T to authenticate.

Retrieving subscriptions for the selection...

[Tenant and subscription selection]

No      Subscription name      Subscription ID      Tenant
-----
[1] *   Spektra EA - AVD   6d7e0652-b03d-4ed2-bf86-f1999cecd17
[2]     Vanguard Class Training 4dd60a42-8e1f-4d8b-b07a-fae39d4a2973

The default is marked with an *; the default tenant is '6d7e0652-b03d-4ed2-bf86-f1999cecd17' and subscription is 'Spektra EA - AVD' (e7f4607f-9595-4b4f-b423-275332b2ed22).

Select a subscription and tenant (Type a number or Enter for no changes): 2

Tenant: 6d7e0652-b03d-4ed2-bf86-f1999cecd17
Subscription: Vanguard Class Training (4dd60a42-8e1f-4d8b-b07a-fae39d4a2973)

[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configuration at https://go.microsoft.com/fwlink/?linkid=271236

If you encounter any problem, please open an issue at https://aka.ms/azclibug

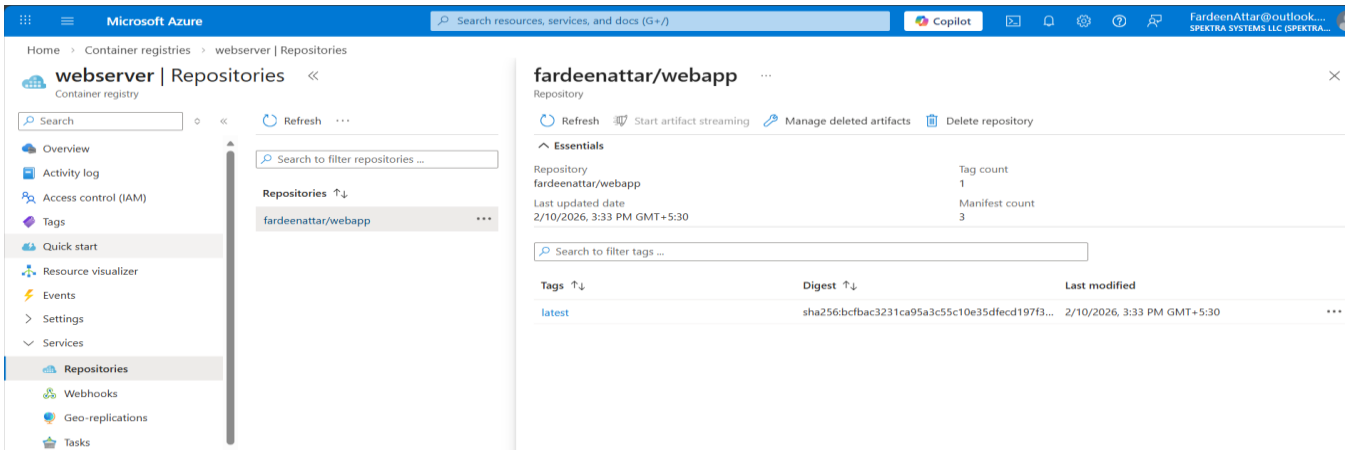
[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.

demo@demovm-docker:~$ az acr login --name webserver
Login Succeeded
demo@demovm-docker:~$ docker images

demo@demovm-docker:~$ az acr login --name webserver
Login Succeeded
demo@demovm-docker:~$ docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
-----                                -                                -            -            -
fardeenattar/webapp:latest           bcfbac3231ca                     237MB       62.9MB       U
nginx:latest                         341bf0f3ce6c                     240MB       65.8MB       U
webapp:latest                        bcfbac3231ca                     237MB       62.9MB       U
demo@demovm-docker:~$ docker tag fardeenattar/webapp:latest webserver.azurecr.io/fardeenattar/webapp:latest
demo@demovm-docker:~$ docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
-----                                -                                -            -            -
fardeenattar/webapp:latest           bcfbac3231ca                     237MB       62.9MB       U
nginx:latest                         341bf0f3ce6c                     240MB       65.8MB       U
webapp:latest                        bcfbac3231ca                     237MB       62.9MB       U
webserver.azurecr.io/fardeenattar/webapp:latest bcfbac3231ca                     237MB       62.9MB       U
demo@demovm-docker:~$ docker push webserver.azurecr.io/fardeenattar/webapp:latest
The push refers to repository [webserver.azurecr.io/fardeenattar/webapp]
46bf3a120c8e: Pushed
f73400a233fd: Pushed
a96de523845a: Pushed
4f4efe02d542: Pushed
7b6cb8ccac7b: Pushed
47cd406a84ef: Pushed
bae5a1799a80: Pushed
33be21097423: Pushed
0c8d55a45c0d: Pushed
latest: digest: sha256:bcfbac3231ca95a3c55c10e35dfecd197f31860a239593c3357c2ae14e1f0351 size: 856
demo@demovm-docker:~$ az acr repository list --name webserver --output table
Result
-----
fardeenattar/webapp
demo@demovm-docker:~$
```



TASK 4 : CREATE AZURE CONTAINER INSTANCE FROM THE IMAGE STORE IN ACR

- Created container instance and accessed via its IP address

